Language, Cognition and Computation (HS 5652)

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What is cognition?

- Cognition refers to a range of mental processes relating to the acquisition, storage, manipulation, and retrieval of information.
- Defined as ‘the mental action or process of acquiring knowledge and understanding through thought, experience, and the senses.’
- Cognition can be separated into multiple distinct functions, dependent on particular brain circuits and neuromodulators.
- We need cognition to help us understand information about the world around us and interact safely with our environment, as the sensory information we receive is vast and complicated: cognition is needed to distill all this information down to its essentials.

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Roles of cognition

- Cognition has a physical basis in the brain with over 100 billion nerve cells in a healthy human brain.
- Each of these can have up to 10,000 connections with other nerve cells called neurons.
- Cognition fundamentally controls our thoughts and behaviors and these are regulated by discrete brain circuits which are underpinned by a number of neurotransmitter systems.
- Brain chemicals that play major roles in regulating cognitive process: dopamine, noradrenaline (norepinephrine), serotonin, acetylcholine, glutamate and GABA

Language in cognition

- Aspects of cognition: Ontological and Logico-Mathematical aspect
- Ontological aspect: the trivial observation that when we speak we speak about something or else we do not speak.
- Questions:

1. Do we only speak about things that really exist in the actual world, or do we also speak about things that do not exist in the actual world but are merely thought-up, either as thought-up individuals or as abstractions or reifications?
2. if we take the view that we also speak about nonexistent things, what sort of things are things that do not exist?

Language in cognition

- In Chomskyan paradigm, the cognition studies deal with the nature of our knowledge of language.
- Linguistic knowledge is a set of such principles and putting that knowledge to use involves the interaction of these principles with other systems of mental representation and computation.
- The data of linguistic performance are a product of our knowledge (competence), but not of the mental system alone.
- I-Language vs. E-Language

Language in cognition

- To understand whether it is form or meaning or the relation between them that is projected through and within human mind.
- Is linguistic form actually projected in the mind?
- Are meaning and relation mentally represented?

Ans: YES (grammatical competence)

- Independent cognitive system of the human mind dedicated to language (I-Langauge)
- Consists of a collection of descriptive statements of language and/or speech events, set of linguistic behaviors and responses.

Neural networks in Language and cognition

- What is the role of language in cognition?
- Do we think with words, or do we use made-up decisions?
- Chomsky (1965): Identification of the first mysteries about language that science had to resolve
- Surrounding language cultures do not carry enough information for a child to learn a language, unless specific learning mechanisms are inborn in the mind of every human being.
- Specific enough for learning complex languages and flexible enough so that the child of any ethnicity can learn her own language.

Neural networks in Language and Cognition

- Syntax is important and language learning is independent of cognition
- Meanings appear independently from language
- Disagreement to this theory emerged in 1970s in the form of cognitive linguistics
- Unification of language and cognition for the creation of meanings
- Knowledge of language is no different from the rest of cognition
- KoL is based on conceptual mechanisms
- Embodied and situated in the environment

Neural networks in Language and Cognition

- Language is not compositional, not all phrases are constructed with words using the same syntax rules and maintaining the same rules (e.g. metaphors)
- Cognitive linguistics is yet to lead to computational linguistic theory to explain how meanings are created.
- Formal apparatus of cognitive linguistics is dominated by logic.
- Evolutionary linguistics emphasized language evolved together with meanings.
- Fundamental property of language: Gets transferred from generation to generation and language mechanisms are shaped by this process.

Neural networks in Language and Cognition

- All linguistic theories are formulated as logical systems and face combinational flexibility
- Do we use phrases to label situations that we have already understood?
- Do we just talk without understanding any cognitive meanings?
- Does difference in cognitive and linguistic abilities have to play any role in to understand language?
- What are the neural mechanisms that enable this flexibility?
- If there is no specific language module (as argued by cognitive linguists), why kids learn language at a young age, but do not think like adults?

Neural networks in Language and Cognition

- Claims:

1. Human language and human cognition are inextricably linked
2. They have evolved jointly.
3. Language and cognition are separate and closely related mechanisms of mind
4. Thinking by using cognitive contexts is possible (Example: playing chess, solving mathematical problems).
5. Language and cognition are embodied

Language in Children

- Language use in children is tacit knowledge.
- Children know more than they say.
- Elimination of closed class or function words (preposition and conjunction etc) and in favor of open class or content words (nouns, verbs etc.)

Language in Children

- Case study:

E.g. Imagine a young child, about 1 year old, interacting with her mother. Typically, children around this age produce one word at a time. When the mother leaves the room and then returns with the child's favorite doll, the child says "doll", not "mother". Later, when the mother is helping her with lunch, the child points at the milk and says "more". Still later, when the child is struggling with her shoes and the mother asks her what she is doing, the simple response is "off".

Ten Lectures on Language as Cognition A Multi-Method Approach

By Dagmar Divjak & Petar Milin

Lecture 1: Usage-Based Linguistics

Aim of the Lecture Series:

- Introduces the “Out of our Minds” research programme by Dagmar Divjak and Petar Milin.
- Funded by the Leverhulme Trust.

Three core goals:

- Understand language as it is known by ordinary speakers, not linguists.
- Understand learning, especially how humans acquire complex systems.
- Improve second/foreign language learning using insights from language and learning research.

Language and Cognition

- Language draws on general cognitive abilities such as:
  i. memory
  ii. Learning
  iii. Categorisation
  iv. attention
- Linguistic knowledge is shaped by experience, not only by abstract rules.

Critique of Traditional Approaches

- Rule-based and introspection-driven models:
  i. assume categorical grammatical rules
  ii. underplay variation and probability

- The authors argue these models cannot fully explain (issues with this approach):
  i. gradient patterns
  ii. frequency effects
  iii. language change

Usage-based linguistics claims

- Grammar emerges from **repeated language use**.
- Linguistic knowledge is built bottom-up from experience.
- There is no need for an innate Universal Grammar.
- Hence, they explain grammar as:
  i. Dynamic
  ii. Experience-based
  (Sensitive to frequency and distribution)

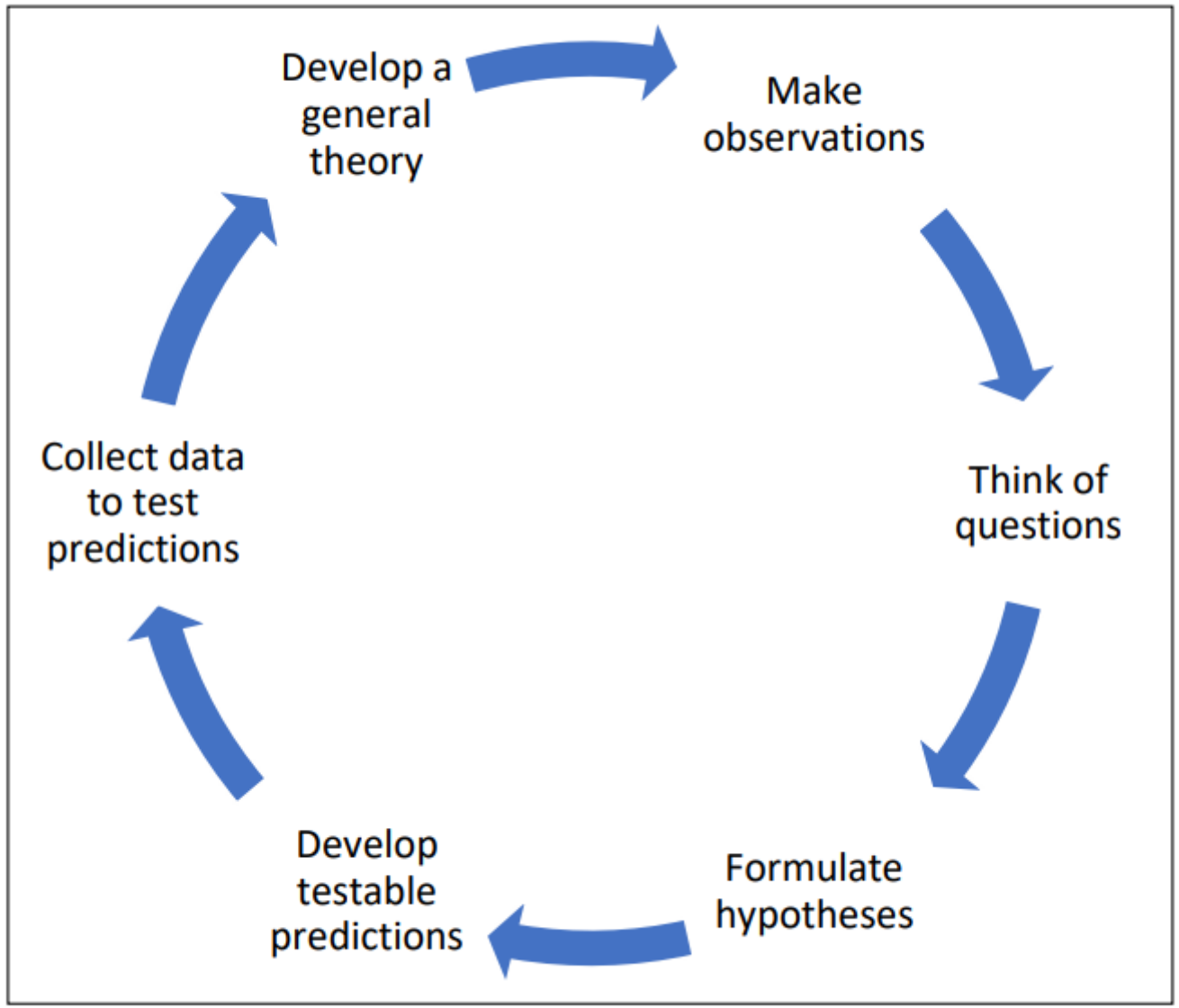


FIGURE 1.2 The scientific cycle

- Shows the scientific cycle of observation, hypothesis, prediction, and testing—grounding how usage-based linguistics builds and evaluates theory. (Fig. 1.2)
- Cognitive linguistics is explicitly usage-based (Langacker, 1987).
- Three foundational assumptions:
  - Linguistic abilities are rooted in general cognition.
  - Grammar is shaped by usage.
  - Meaning is similar to conceptualisation, not truth conditions.
- Early focus in cognitive linguistics were:
  - Categorisation
  - Generalisation
  - Abstraction

- The Quantitative Turn in Cognitive Linguistics
- Early cognitive linguistics relied heavily on introspection.
- Early 2000s: major methodological shift toward empirical data.

Adoption of:

  - Corpora
  - Quantitative and statistical methods
  - Around 2013, most cognitive linguistic research involved data analysis.

Reason for success:

- Usage-based theory fits naturally with corpus data, which was considered an **ideal approach.**

- Emerging issue: Are corpus patterns cognitively real?
- Corpus methods easily detect statistical patterns.
- Key question:

  "Do these patterns exist only in the data, or also in speakers' minds?"
- Risk:

  : Linguists may find patterns that are descriptively accurate but psychologically irrelevant.

This leads to a central tension about **descriptive accuracy** vs. **psychological reality**

The Cognitive Commitment:

- Introduced by Lakoff (1990)
- Core idea:
  i. Linguistic theories must accord with what is known about the mind and brain.
- Can be interpreted in three ways:
  i. Cognitive plausibility (motivations from human cognition)
  ii. Psychological reality (mental representations and processing)
  iii. Neurobiological reality (brain evidence – still rare)
- Cognitive linguistics has become theoretically and methodologically diverse, sometimes inconsistent.

Corpus Linguistics: Three Interpretations

1. Corpus linguistics as a theory (corpus-driven)
    - Meaning is in discourse, not minds.
    - Minimal assumptions.
    - Incompatible with cognitive linguistics.
    - Marginal and impractical.
2. Corpus linguistics as a discipline
    - Has terminology and accumulated knowledge.
    - Methods are flexible, not obligatory.
    - Empirical but methodologically loose.
3. Corpus linguistics as a methodology
    - Uses corpora to test theory-driven hypotheses
    - Allows predefined categories such as verbs
    - Corpus linguistics as a methodology
    - Uses corpora to test theory-driven hypotheses.
    - Allows predefined categories (verbs, constructions, etc.).

- Raw data alone are not meaningful
- Linguistics insight arises from:

i. Systematic analysis

ii. Theory-guided interpretation

- Can be understood with the help of the figure depicting:



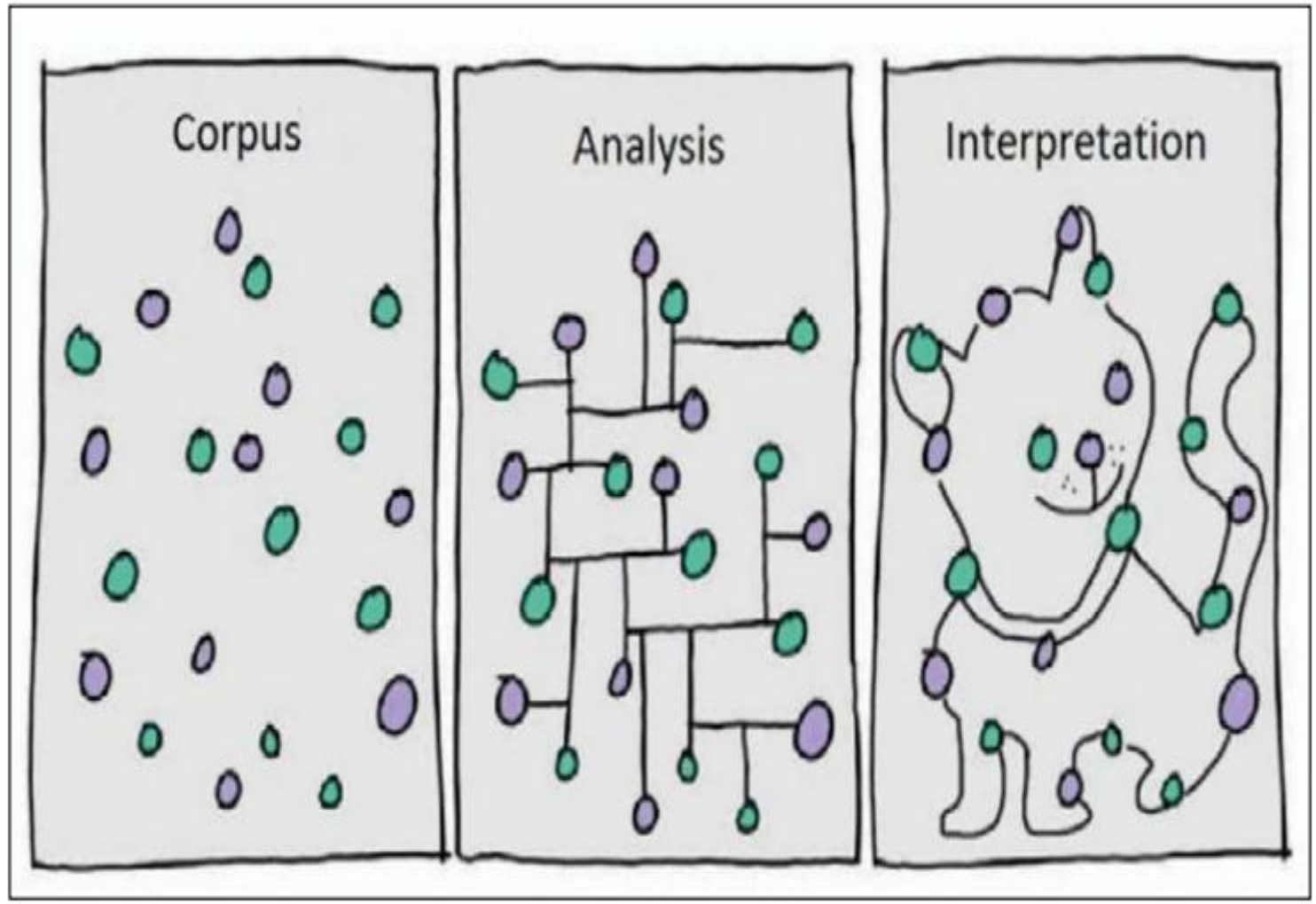


FIGURE 1.5 Data – Analysis – Interpretation

Combining Corpus Linguistics and Cognitive Linguistics

- Cognitive linguistics provides:
  - i. Theory
  - ii. Hypotheses
- Corpus methods provide:
  - i. Data
  - ii. Testing ground
- Key assumption: Speakers choose different forms because they serve different communicative functions.

The Multi-Method Approach

1. Corpus Linguistics:

   : Reveals patterns in **natural language use**

   : Shows frequency and variation

2. Experiments:

   : test speaker knowledge

   : gives controlled evidence

3. Statistical Modelling:

   : quantifies effects

   : identifies significant predictors

Limitations of Corpus and Experimental Validation

- Corpora:
  i. Snapshots of usage
  ii. Often written, not spoken
  iii. Not individual experience
- Experiments:
  i. Often artificial
  ii. Sometimes use unnatural sentences
- Validation should be bidirectional:
  - Corpus findings should be tested experimentally.
  - Experiments should be checked against real usage.

Interdisciplinary Collaboration in Language Research

Requires collaboration between:
i. Linguists: Study the structure, meaning, and diversity of language
ii. Psychologists: Examine human language processing and behaviour
iii. Computational modellers: Build formal models to test predictions

Shared requirements:
i. Interest in Language: Curiosity across disciplines
ii. Commitment to the Empirical Cycle: Observe → Hypothesise → Test → Revise
iii. Partial Understanding of Each Other's Methods: Enables effective communication and integration

Lecture 2: Memory and Attention

Purpose of the Lecture:

- Examines how memory and attention shape language learning and representation.
- Argues that linguistic theory must respect well-established cognitive constraints.
- Moves cognitive linguistics closer to psychological realism, not just descriptive adequacy.

Why Memory and Attention Matter for Language

- Language learning depends on:
  i. What learners notice
  ii. What they remember
- Not all input is equally available for learning.
- Hence, frequency alone is insufficient to explain linguistic knowledge.

Claim: Linguistic structure reflects what learners can attend to and store, not just what occurs in input.

Types of Memory Relevant to Language

Memory System	Main Function	Key Properties	Consequences for Learning
Working Memory	Supports online processing	i. Handles moment-to-moment understanding ii. Limited capacity iii. Constrains how many cues can be integrated	Influences real-time comprehension and processing efficiency
Long-Term Memory	Stores lexical & grammatical knowledge	i. Builds through repetition & exposure ii. Durable representations iii. Sensitive to frequency	Frequent items gain stronger retrieval strength and become entrenched

Memory Is Not a Perfect Recorder

- Memory is noisy, lossy and selective
- Learners do not store all instances equally.
- Consequence:

  : Linguistic knowledge is shaped by biased sampling of experience.

- These challenges:

i. Rule-based, abstract models

ii. Assumptions of complete input storage

The Role of Attention

- Attention determines which aspects of input are encoded.
- Influenced by:
  i. Salience
  ii. Task relevance
  iii. Expectations
- Attention is:
  i. Limited
  ii. Competitive

Key implication: If a learner does not attend to a feature, it is unlikely to be learned.

Frequency Revisited: Beyond Raw Counts

- Traditional usage-based linguistics:
  : Emphasised raw frequency.
- Authors argue for:
  : Experience-weighted frequency
- Important distinctions:
  : Token frequency vs. type frequency
  : Recent vs. distant experience
  : Attended vs. unattended input
- Learning depends on what is noticed, retained and retrieved.

Entrenchment and Forgetting

- Entrenchment: Repeated exposure strengthens representations.
- Forgetting: Memory traces decay over time.
- Learning is a balance between:
  : Reinforcement
  : Decay
- This leads to:
  : Gradient representations
  : Probabilistic grammatical knowledge

Individual Differences

- Learners differ in:
  : Memory capacity
  : Attentional control
- These differences explain:
  : Variation in language proficiency
  : Different generalisations from similar input

Implication: Linguistic theory must accommodate variation, not just idealised speakers.

Implications for Linguistic Description

- Linguistic categories should:
  : Reflect memory-based representations
  : Avoid unnecessary abstraction
- Overly complex models may:
  : Fit data well
  : But exceed cognitive capacity
- Therefore, cognitive realism should guide annotation and modelling choices.

Implications for Modelling Language Learning

- Models should:
  : Include memory limitations
  : Incorporate attentional constraints
- Simple learning mechanisms can:
  : Produce complex linguistic patterns
- Rejects:
  : Models assuming unlimited memory or attention

To summarise:

- Memory and attention are central mechanisms in language learning.
- Language structure reflects:
  : Cognitive constraints
  : Selective experience
- Usage-based models must:
  : Go beyond frequency
  : Be grounded in realistic learning conditions

Lecture 3: Construal: A Visual World Approach to Constructional Alternations

Aim of the chapter:

- Explores construal: how speakers choose different linguistic forms to express the same situation.
- Focuses on constructional alternations (different grammatical options).
- Uses the Visual World Paradigm and corpus data to connect:
  i. Usage
  ii. Attention
  iii. Cognitive processing

What Is Construal?

- Construal is understood as how a speaker mentally represents a situation.
- The same event can be:
  i. Conceptualised differently
  ii. Expressed using different constructions

Examples:

- Active vs. passive
- Dative alternation (NP vs. PP)
- Word order variation
- Which means that linguistic choices reflect perspective, not different meanings in the world (Fig. 3.1)

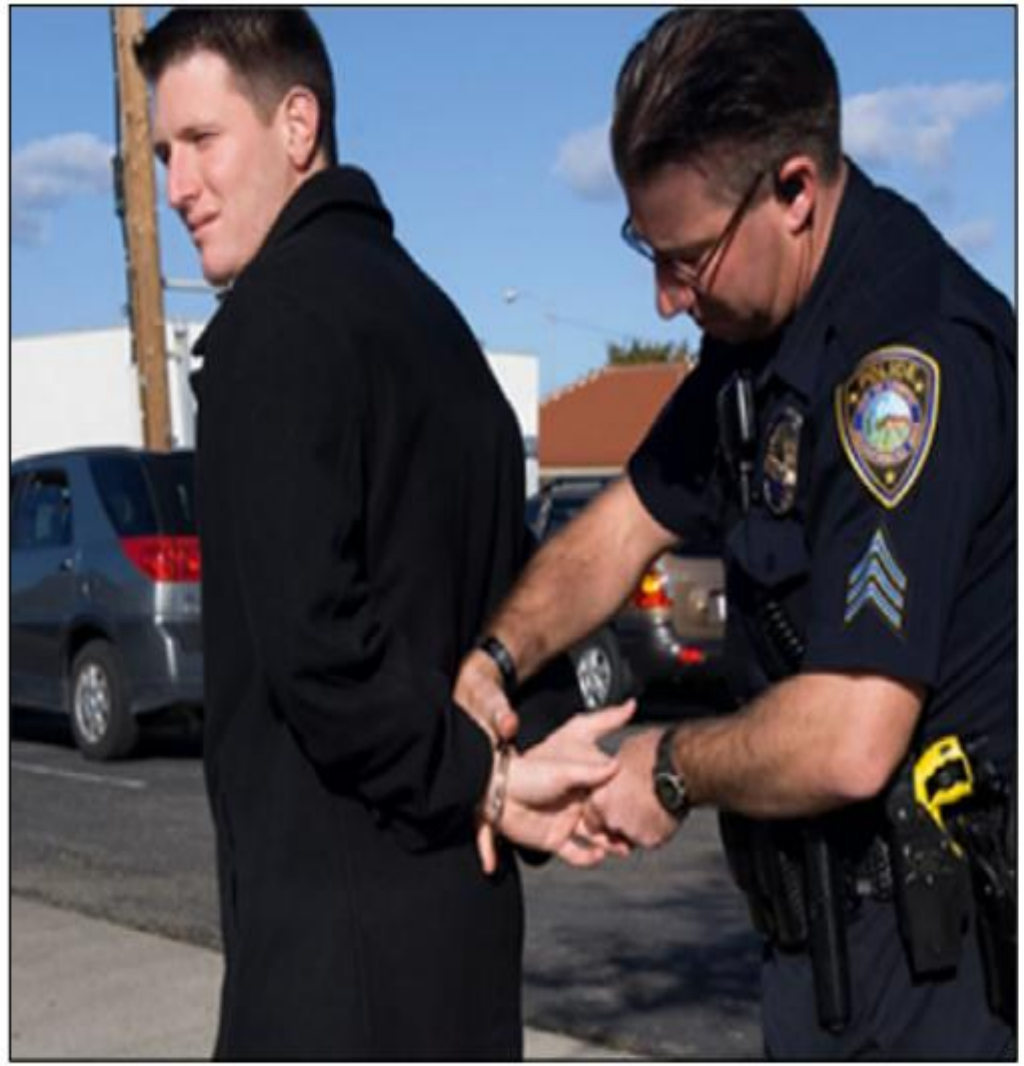


FIGURE 3.1
Example picture co-construed by active or passive sentence

Tomlin's Fish Film

- Visual cue: biased active/passive selection
- Shows **attention → grammar** link during sentence production
- Provides conceptual motivation for **manipulating linguistic construal**

in eye-tracking studies

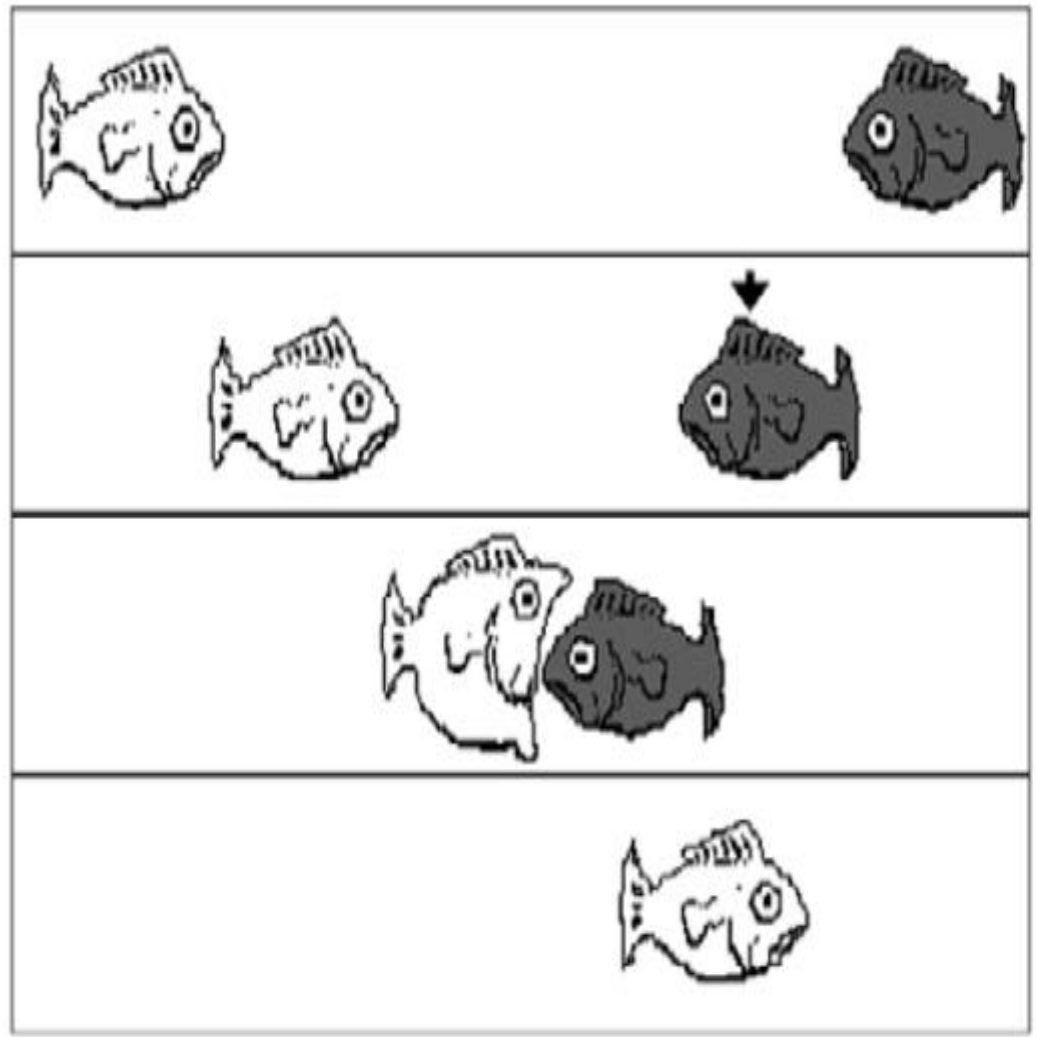


FIGURE 3.4
A screenshot from
Tomlin's Fish Film

Construal in Cognitive Linguistics

- Cognitive linguistics assumes:
  : Meaning is not objective truth
  : Meaning is similar to conceptualisation by the speaker
- Construal involves choices about:
  : Focus and backgrounding
  : Prominence
  : Perspective
  : Level of detail

Experimental Design Diagram (Construal Types)

- Three alternations tested:

1. **Prepositions:** typical vs. atypical
   (e.g., *on the table* vs. *above the cup*)
2. **Voice:** active vs. passive
   (*Ram killed Raavan* vs. *Raavan was killed by Ram*)
3. **Dative:** NP- dative
   (*John gave Mary…*) vs. PP-dative (*John gave to Mary…*)

Constructional Alternations

- There were alternative instances where:
  : Multiple constructions express similar content
  : Choice is not random
- Some examples (Active vs. Passive) :
  i. *Give Mary the book* vs. *Give the book to Mary*
  ii. *The chef cooked the meal* vs. *The meal was cooked*
- Traditional problem:

  Why do speakers choose one construction over another?

Limits of Purely Corpus-Based Explanations

- Corpus models identify:
  : Predictors (animacy, length, givenness, etc.)
- There was an issue with the model that says that:

  Statistical predictors ≠ cognitive explanation
- Corpus models describe what correlates, not how choices are made in real time.

The Visual World Paradigm

- Experimental method from psycholinguistics.
- Tracks:
  : Eye movements
  : Attention during language processing
- Assumption:
  "Where people look reflects what they attend to cognitively"
- Used to test if corpus-derived predictors influence real-time processing.
- Longer durations when integrating **sentence + scene**
- Result: Prepositions show effects on all eye-tracking measures, voice affects only the order of access, and dative shows no significant effects, indicating a cline of construal strength

(Preposition > Voice > Dative)

Measure	Preposition	Voice	Dative
Order of Access	✓	✓	✗
First gaze duration	✓	✗	✗
Overall gaze duration	(✓)	✗	✗

FIGURE 3.15 Summary of attested differences for construal types and across the three eye-tracking measures

Attention as the Missing Link

- Construal is guided by **attention allocation**.
- Speakers choose constructions based on:
  i. What they attend to first
  ii. What is most salient
- Important shift:

  "Construal is not a post-hoc explanation; it happens during processing"

Integrating Corpus Data and Experimental Evidence

Corpus analysis:

: Identifies distributional tendencies

Visual world experiments:

: Test whether these tendencies guide attention

Convergence:

- When corpus predictors align with eye-tracking results, it leads to a stronger cognitive reality.

Rethinking Grammatical Choice

- Construction choice is:
  : Dynamic
  : Incremental
  : Attention-driven
- Speakers do not:
  : Compute abstract rules
  : Choose from a menu of constructions

Instead:

- Choices emerge from competition between attended elements.

Implications for Linguistic Theory

- Supports:
  : Usage-based, processing-driven accounts
- Challenges:
  i. Rule-based models
  ii. Static representations

Encourages:
  : Linking grammar to real-time cognition

To conclude:

- Construal explains why alternations exist.
- Constructional choice reflects:
  - Attention
  - Salience
  - Processing dynamics
- Cognitive reality is strengthened when:

  "Corpus findings align with processing data"

Eye-Tracking Experiment: Design and Key Findings

Experimental Design

- Participants process inflected word forms
- Eye movements recorded:
  - : fixation duration
  - : regressions
- Manipulated factors:
  - : high vs. low frequency forms
  - : regular vs. irregular inflection
  - : strong vs. weak class membership

Key Findings:

- Faster fixation times for:
  : high-frequency forms
  : well-entrenched paradigms
- Longer fixation times for:
  : low-frequency forms
  : irregular or weakly patterned forms

Interpretation:

: Processing difficulty reflects gradient paradigm strength

- Eye-movement patterns mirror:
  : usage frequency
  : degree of abstraction
- Provides real-time evidence that paradigms are psychologically real

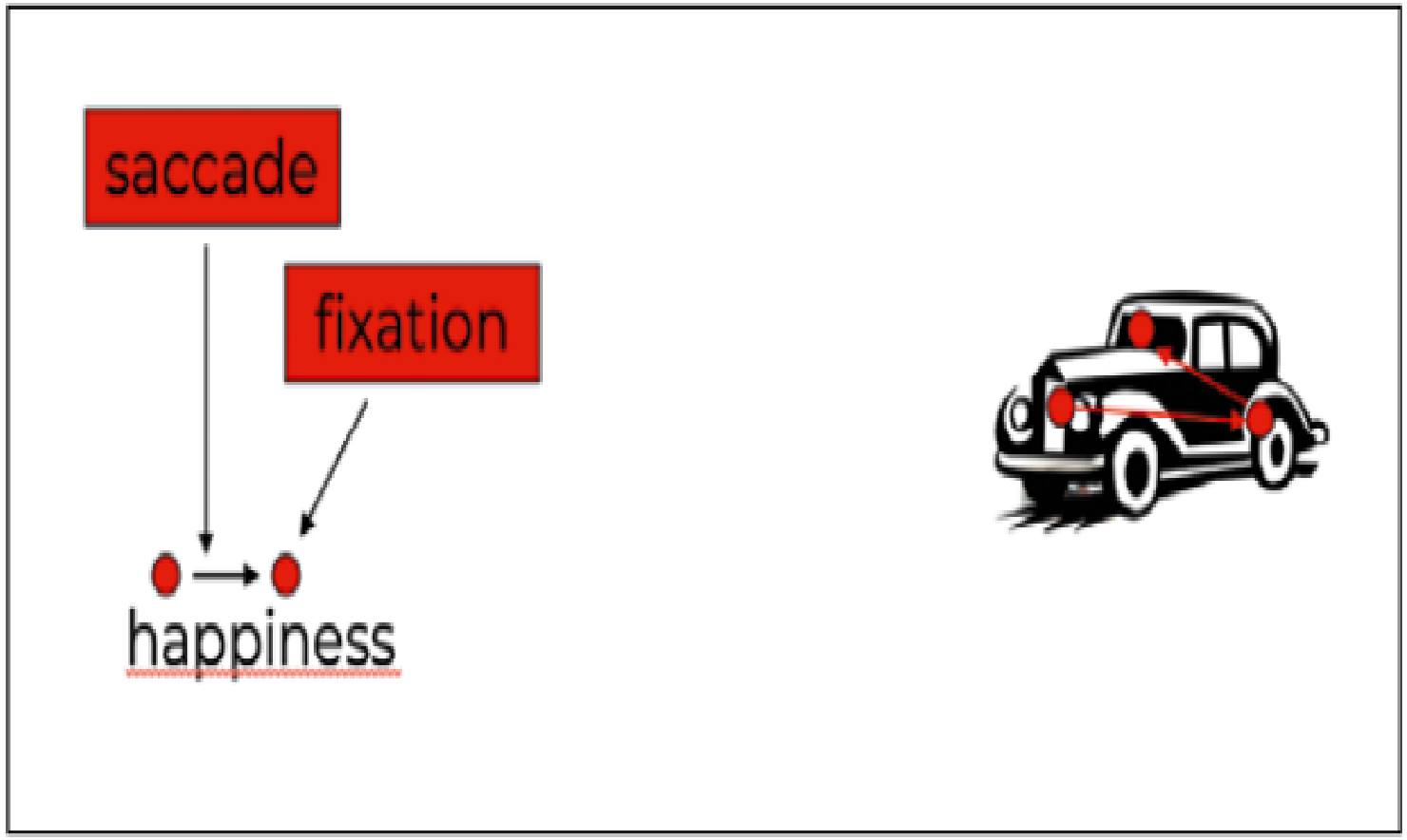


FIGURE 7.4 Saccades and fixations in eye-movements



FIGURE 3.11
Typical fixation positions represented as heat-map



FIGURE 7.8 Example of fixations while reading a simple sentence

Some measures that are often used in reading studies are:

- First Fixation Duration, Second Fixation Duration
- First Fixation Location, Second Fixation Location
- Gaze Duration, Total Fixation Duration
- Number of Fixations



FIGURE 7.12 Example of foveal and parafoveal uptake while reading experimental sentences

Language, Mind and World

- Represents relations that are keys to understanding our place in reality.
- Constitutes the meaningfulness of language
- Mind<--> World
- Mind-→ Language
- Understanding is the cornerstone concept: An account of what constitutes the ability to understand it
- Some philosophers focus more on production than consumption
- Analysis of linguistic meaning in terms of the goals and practices of speaker

Language -> Mind

- If mind assigns meaning to language, so also language *enables* and *channels* mind.
- Acquiring and trafficking in a language brings one concepts, thoughts and habits of thought, with all sorts of consequences
- Language is so crucial to our ability to frame the sophisticated thoughts that appear essential to language-use and understanding.

Language <-> World

- Language is the vehicle of our descriptions and explanations of reality
- The key to meaning is the notion of a *truth-condition*.
- Statement: ‘Chennai is bigger than Madurai.’
- The key to meaning is a notion of *correct use*.
- The Mind/Language and Language/World connections can seem to place a tension on the notion of meaning (meaning is whatever we cognitively *grasp*, but the meaning of language just is its bearing on the world).

Language and Mind

- Language of Thought: A formal language that is postulated to be encoded in the brains of intelligent creatures as a vehicle for their thought (Rey, Georges)
- Question: Does it resemble any ‘natural’ language spoken by anyone?
- Chief function: A medium of representation over which the computations posited by cognitive psychologists are defined
- Features:

i. Productivity, Systematicity, (hyper-) intensionality of the thought

ii. Promiscuity of their attitude

iii. Ability to reason in familiar deductive, inductive and practical ways

Language and Mind

- Language of Thought Hypothesis (LOTH):

i. Thinking occurs in mental language

ii. Mental language is similar with spoken language (combination of words; meaningful phrases and sentences; sentential meaning is systematic)

iii. Claim: LOT wants to construe propositional attitude tokens as relations to symbol tokens.

iv. *P* is to bear a certain relation to a token of symbol which means that *P*.

v. Symbols are intentional contents and their tokens are physical objects

Language and Thought

- Some questions to ponder over:

i. Does the language you speak affect how you think about the world? (Mandarin vs. English speaker's conceptions of time by Boroditsky 2001)
ii. Is language a psychological process?
iii. How do humans acquire basic language skills?
iv. How does language ability help humans in thinking?
v. How and why individuals differ in their ability to learn language?
vi. How does a child learn the grammar of a language?
vii. What is the relationship between written and spoken language?

Language and Thought

- Our speech is based on our thought
- Perception and expression are understood to be synonymous
- What one says is dependent of how it is encoded and decoded in the mind
- Contradictory view: What one perceives is dependent on the spoken word

Language and Thought

Linguistic Relativity (Sapir-Whorf Hypothesis):

i. Language affects its speaker's world view or cognition, thus people's perceptions are relative to their spoken language
ii. Linguistic relativity is distinguished both from simple linguistic diversity and from strict linguistic determinism
iii. It represents language as a mould in terms of which thought categories are cast (Chandler 2002)
iv. Linguistic determinism: Refers to the concept that what is said, has only some effect on how concepts are recognized by the mind
v. Linguistic relativism: "Distinctions encoded in one language are unique to that language along," and that "there is no limit to the structural diversity of languages

- This is the eighth-grade final exam from 1895 in Salina, Kansas, USA. It was taken from the original document on file at the Smokey Valley Genealogical Society and Library in Salina , and reprinted by the Salina Journal.
-

8th Grade Final Exam: Salina, Kansas - 1895

Grammar (Time, one hour)
1. Give nine rules for the use of capital letters.
2. Name the parts of speech and define those that have no modifications.
3. Define verse, stanza and paragraph
4. What are the principal parts of a verb? Give principal parts of 'lie,''play,' and 'run.'
5. Define case; illustrate each case.
6 What is punctuation? Give rules for principal marks of punctuation.
7 - 10. Write a composition of about 150 words and show therein that you understand the practical use of the rules of grammar.

(Courtesy: Prof. LSG’s mail to the faculty at IITM)